

Common-angle migration: A strategy for imaging complex media

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ABSTRACT

Complex velocity models characterized by strong lateral variations are certainly a great motivation, but also a great challenge, for depth imaging. In this context, some unexpected results can occur when using depth imaging algorithms. In general, after a common shot or common offset migration, the resulting depth images are sorted into common-image gathers (CIG), for further processing such as migration-based velocity analysis or amplitude-variation-with-offset analysis. In this paper, we show that CIGs calculated by common-shot or common-offset migration can be strongly affected by artifacts, even when a correct velocity model is used for the migration. The CIGs are simply not flat, due to unexpected curved events (kinematic artifacts) and strong lateral variations of the amplitude (dynamic artifacts). Kinematic artifacts do not depend on the migration algorithm provided it can take into account lateral variations of the velocity model. This can be observed when migrating the 2-D Marmousi dataset either with a wave-equation migration or with a multivalued Kirchhoff migration/inversion. On the contrary, dynamic artifacts are specific to multi-arrival ray-based migration/inversion. This approach, which should provide a quantitative estimation of the reflectivity of the model, provides in this context dramatic results.

In this paper, we propose an analysis of these artifacts through the study of the ray-based migration/inversion

operator. The artifacts appear when migrating a single-fold subdata set with multivalued ray fields. They are due to the ambiguous focusing of individual reflected events at different locations in the image. No information is a priori available in the single-fold data set for selecting the focusing position, while migration of multifold data would provide this information and remove the artifacts by the stack of the CIGs. Analysis of the migration/inversion operator provides a physical condition, the imaging condition, for insuring artifact free CIGs. The specific cases of common-shot and common-offset single-fold gathers are studied. It appears clearly that the imaging condition generally breaks down in complex velocity models for both these configurations. For artifact free CIGs, we propose a novel strategy: compute CIGs versus the diffracting/reflecting angle. Working in the angle domain seems the natural way for unfolding multivalued ray fields, and it can be demonstrated theoretically and practically that common-angle imaging satisfies the imaging condition in the great majority of cases. Practically, the sorting into angle gathers can not be done a priori over the data set, but is done in the inner depth migration loop. Depth-migrated images are obtained for each angle range. A canonical example is used for illustrating the theoretical derivations. Finally, an application to the Marmousi model is presented, demonstrating the relevance of the approach.

INTRODUCTION

In prestack seismic processing, common-image gathers (CIGs) provided by common-offset (or common-shot) depth migration are now commonly used for migration-based velocity analysis (Jin and Madariaga, 1993, 1994; Symes, 1993)

or amplitude-variation-with-offset (AVO) analysis (Beydoun et al., 1993; Tura et al., 1998). There is a general agreement that common offset gathers are the convenient minimum data sets for such analysis (Ehinger et al., 1996). When working in the complex media, artifacts appear in the CIG: Duquet (1996) observed in the 2-D Marmousi dataset (Bourgeois

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et al., 1991) that, curiously for the exact velocity macromodel, CIGs in offset were not flat although the migration stack was satisfactory. This observation gave a pounding to the well-admitted principle of migration-based velocity analysis, and Duquet (1996) proposed to introduce an extra constrain in the migration to enforce the flatness of the events.

All these results were obtained by wave-equation migration that does not allow a theoretical analysis of the problem. On the contrary, asymptotic imaging (or ray-based imaging) offers a powerful theoretical framework to study such problems.

Many theoretical works have been published on asymptotic imaging. The famous paper of Beylkin (1985) has been the basis of quantitative asymptotic imaging (Bleistein, 1987; Beylkin and Burridge, 1990; Jin et al., 1992). It is what we call “migration/inversion.” Ray-based migration also provides opportunities for resolution analysis (Operto et al., 1998) or conditioning analysis of multiparameter migration/inversion (Jin et al., 1992; Forgues and Lambaré, 1997). The case of triplicated ray fields was not taken into account by the initial work of Beylkin (1985), and the problem was later addressed by Rakesh (1988), ten Kroode et al. (1994, 1998), and Nolan and Symes (1996). These theoretical papers demonstrated that asymptotic imaging could be extended to the case of multivalued ray fields provided that some conditions were satisfied such as the travel-time injectivity condition (TIC) introduced by ten Kroode et al. (1994, 1998) or the imaging condition introduced by Nolan and Symes (1996) as a generalization of the TIC. Whereas the work by ten Kroode et al. (1994, 1998) was limited to the case of ideal 2-D multifold data, Nolan and Symes (1996) addressed the problem of migration of data collected with real 3-D acquisition geometries. Those authors noticed that common-shot or common-offset migration may infringe on the imaging condition and that, consequently, corresponding images could be affected by significant artifacts. Those authors illustrated such artifacts on canonical synthetic data and suggested some procedures for reducing the artifacts.

The problems encountered when running common-offset migration in case of triplicated ray fields are not very well known in the geophysical community. In case of real data, the artifacts in CIGs obtained by Duquet (1996) are buried under the problem of the estimation of complex velocity macromodels, but also under the problem of the numerical implementation of the migration with multivalued traveltimes maps. Another reason for this ignorance is the lack of convincing illustrations of the failure of common-offset imaging in case of triplicated media. It is precisely one of the goals of this paper to provide such an illustration. We also go further and propose a simple and efficient strategy for overcoming these artifacts: common-diffracting-angle migration, which had been formerly proposed by de Hoop et al. (1994) for migration/inversion in anisotropic media.

Papers by ten Kroode et al. (1994, 1998), Nolan and Symes (1996), and Nolan (1996) require a significant mathematical background on Fourier integral operators and pseudodifferential operators (Treves, 1980). Such developments are simply out of the scope of our paper. We rather try to present the theoretical results in a physical way and to illustrate them with convincing examples.

In this paper, we first recast the theoretical considerations of ten Kroode et al. (1994, 1998), Nolan and Symes, (1996) and

Nolan (1996) into a physical analysis of migration/inversion (Beylkin, 1985; Bleistein, 1987; Jin et al., 1992) of single-fold data. The imaging condition, required to obtain artifact free CIGs, is derived from a physical analysis of migration/inversion schemes. Migration/inversion formulas are established for the Born and Kirchhoff approaches. Common-shot and common-offset cases are analyzed in view of the imaging condition, revealing the general failure of migration for such configurations. As a solution for obtaining CIGs, common-diffracting-angle migration/inversion is proposed. A canonical test is used to illustrate the phenomena and, finally, we present an application on the synthetic 2-D Marmousi dataset (Bourgeois et al., 1991).

ASYMPTOTIC LINEAR FORWARD MODELING

Migration can be recast in the general frame of asymptotic linearized seismic inverse theory. Born or Kirchhoff approximations can be used for the linear forward modeling, depending on the description of model as a reflectivity function or as a model perturbation (typically the impedances).

Ray + Born approximation

Born approximation is the linear approximation of the relation connecting the perturbations of data $\delta\mathcal{G}(\mathbf{r}, \omega; \mathbf{s})$ ($\mathbf{s}$ and $\mathbf{r}$ denote the shot and receiver positions and ω the angular frequency) to the perturbations of model $\delta m(\mathbf{x})$ ($\mathbf{x}$ denotes the position in the model) around a reference model $m_0(\mathbf{x})$ and a reference data $\mathcal{G}_0(\mathbf{r}, \omega; \mathbf{s})$. Validity of the ray + Born approximation requires the reference model to contain the long-wavelength components of the exact model and that only single diffractions/reflections exist. The first-order Born operator requires the computation of Green's functions in the reference model. When they are estimated by ray theory, we get the ray + Born approximation.

Consider the scalar wave equation with a perturbation of the square slowness of the reference model $1/c_0^2(\mathbf{x})$ (Thierry et al., 1999b). Ray + Born approximation can be expressed in the frequency domain [see Thierry et al. (1999b) for the convention in Fourier transforms] as

$$\delta\mathcal{G}(\mathbf{r}, \omega; \mathbf{s}) \cong \mathcal{B}(\mathbf{r}, \omega, \mathbf{s}) [\delta m(\mathbf{x})] \quad (1)$$

$$\cong \int d\mathbf{x} \delta m(\mathbf{x}) \sum_{n=1}^{N(\mathbf{s}, \mathbf{x}, \mathbf{r})} \mathcal{B}_n(\mathbf{r}, \omega, \mathbf{s}; \mathbf{x}), \quad (2)$$

where

$$\mathcal{B}_n(\mathbf{r}, \omega, \mathbf{s}; \mathbf{x}) = \mathcal{K}(\omega) \mathcal{A}_n(\mathbf{r}, \mathbf{x}, \mathbf{s}) e^{i\omega T_n(\mathbf{r}, \mathbf{x}, \mathbf{s})}, \quad (3)$$

where $\mathcal{K}$, $\mathcal{A}$, and T denote, respectively, the signature, the amplitude, and the phase of the Born operator associated with the receiver and source ray branch n . N is the total number of branches of the ray + Born operator (i.e., the number of receiver ray branches multiplied by the number of source ray branches). We have for each contribution $\mathcal{B}_n$

$$\begin{cases} \mathcal{A}(\mathbf{r}, \mathbf{x}, \mathbf{s}) = \mathcal{A}(\mathbf{r}, \mathbf{x}) \mathcal{A}(\mathbf{x}, \mathbf{s}) \\ \mathcal{K}(\omega) = \omega^2 S^2(\omega) \\ T(\mathbf{r}, \mathbf{x}, \mathbf{s}) = T(\mathbf{r}, \mathbf{x}) + T(\mathbf{x}, \mathbf{s}) \\ \quad - \frac{\pi}{2} \frac{\text{sign}(\omega)}{\omega} (\alpha(\mathbf{r}, \mathbf{x}) + \alpha(\mathbf{x}, \mathbf{s})), \end{cases} \quad (4)$$

where A denotes the amplitude, T the traveltime, S the signature of the Green function, and α the KMAH index (Chapman, 1985; Operto et al., 2000).

In two dimensions, S and A are given by

$$S(\omega) = \frac{1}{\sqrt{-i\omega}} \quad \text{and} \quad A(\mathbf{x}, \mathbf{s}) = \sqrt{\frac{c_0(\mathbf{x})}{8\pi |\mathcal{J}(\mathbf{x}, \mathbf{s})|}}, \quad (5)$$

where $\mathcal{J}(\mathbf{r}, \mathbf{s})$ denotes the geometrical spreading [see Thierry et al. (1999b) for the precise definition] associated with the 2-D asymptotic Green function.

Ray + Kirchhoff approximation

Kirchhoff approximation is also an asymptotic linear approximation of the relation connecting the perturbations of the data to the perturbations of the model. The perturbed model is however not described by a square slowness perturbation as in the Born approximation, but by a distribution of specular reflection angle $\theta(\mathbf{x})$, defined by the given acquisition geometry (common-shot or common-offset data sets) and by the associated reflectivity function $R(\mathbf{x}, \theta(\mathbf{r}, \mathbf{x}, \mathbf{s}))$ (Figure 1). Since Kirchhoff approximation is based on the concept of specular reflection, it is an asymptotic approximation consistent with the use of asymptotic Green's functions. The first-order asymptotic Kirchhoff approximation is given by

$$\delta\mathcal{G}(\mathbf{r}, \omega; \mathbf{s}) = \frac{1}{-i\omega} \int d\mathbf{x} \sum_{n=1}^{N(\mathbf{s}, \mathbf{x}, \mathbf{r})} R(\mathbf{x}, \theta_n(\mathbf{r}, \mathbf{x}, \mathbf{s})) \times |\mathbf{q}_n(\mathbf{r}, \mathbf{x}, \mathbf{s})| \mathcal{B}_n(\mathbf{r}, \omega, \mathbf{s}; \mathbf{x}), \quad (6)$$

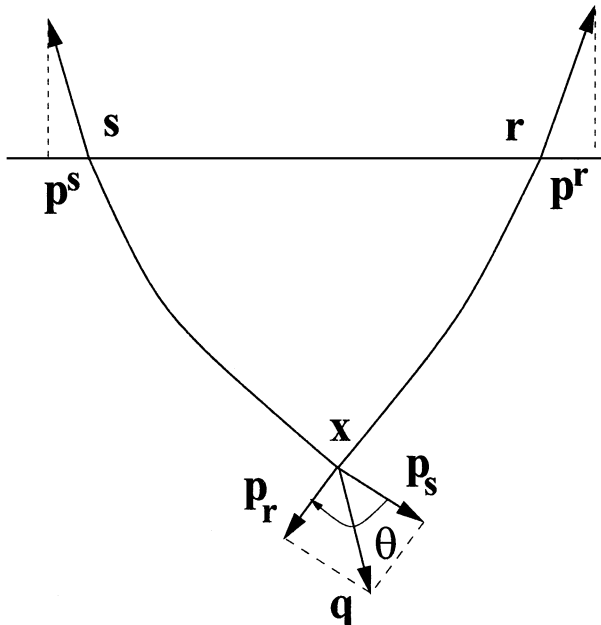


FIG. 1. Some ray parameters involved in migration/inversion: θ denotes the diffracting/reflecting angle; p^s and p^r denote the horizontal component of the slowness vectors at the surface, whereas $\mathbf{p}_s$ and $\mathbf{p}_r$ are the slowness vectors at the position $\mathbf{x}$ in the background.

where $\mathbf{q} = \mathbf{p}_s + \mathbf{p}_r$ is the sum of the two slowness vectors at the scattering point $\mathbf{x}$ (Figure 1) [with $|\mathbf{q}| = 2 \cos(\theta/2)/c_0$], and $\mathcal{B}_n$ is the kernel of the folded ray + Born operator [equation (3)].

MIGRATION/INVERSION OF SINGLE-FOLD DATA SETS WITH MULTIPLE ARRIVALS

Migration/inversion theory

For a given data set, the ray + Born or ray + Kirchhoff approximations can be inverted within the general framework of inverse problem theory (Tarantola, 1987). Rather than the full multifold data parameterized by $(\mathbf{r}, \omega, \mathbf{s})$, we consider individual subsets such as common-shot or common-offset gathers. Such single-fold subsets are often supposed to give consistent individual images of the model. The multifold data set $(\mathbf{s}, \mathbf{r}, \omega)$ is split into a collection of single-fold data sets parameterized by (γ, σ, ω) ; $\gamma = \text{constant}$ defines a single-fold subset (common-shot or common-offset gathers) parameterized by σ in the space of traces positions $(\mathbf{r}, \mathbf{s})$ (for example, the receiver or the midpoint position, respectively, in case of common-shot or common-offset gathers).

Consider a single-fold data set γ_0 . Following Thierry et al. (1999b), we take a weighted ℓ^2 cost function,

$$\mathcal{C}[\delta m|_{\gamma_0}] = \frac{1}{2} \int d\omega \int_{\gamma_0} d\sigma \mathcal{Q} |\delta\mathcal{G}_{\text{obs}}(\mathbf{r}(\sigma), \omega; \mathbf{s}(\sigma)) - \delta\mathcal{G}_{\text{cal}}(\mathbf{r}(\sigma), \omega; \mathbf{s}(\sigma))|^2, \quad (7)$$

where $\delta\mathcal{G}_{\text{obs}}$ are the observed data. Weighting factor $\mathcal{Q}$ is equivalent to the inverse covariance matrix in the data space of the inverse problem theory (Tarantola, 1987). In our approach, as we will see later on, even if $\mathcal{Q}$ keeps some physical interpretation, it is introduced for numerical reasons (simplification of the Hessian operator).

The expression of the solution δm minimizing the cost function is given by

$$\delta m = (\mathcal{B}^\dagger \mathcal{Q} \mathcal{B})^{-1} \mathcal{B}^\dagger \mathcal{Q} \delta\mathcal{G}_{\text{obs}}, \quad (8)$$

where † denotes the adjoint operator; $-\mathcal{B}^\dagger \mathcal{Q} \delta\mathcal{G}_{\text{obs}}$ is the gradient of the cost function and the operator, $\mathcal{B}^\dagger \mathcal{Q} \mathcal{B}$, is the Hessian. For usual seismic applications, the Hessian is a very large matrix difficult to invert numerically, and local iterative gradient-type minimization algorithms were proposed (Beydoun and Mendes, 1989). In order to accelerate the convergence of the algorithm, Jin et al. (1992) proposed to introduce a weight, $\mathcal{Q}$, such that the Hessian matrix is asymptotically diagonal (i.e. easily invertible). Jin et al.'s method allowed reconciliation of the direct inversion approach of Beylkin (Beylkin, 1985; Bleistein, 1987) with the stochastic formulation of inverse problems theory (Tarantola, 1987). The following developments are based on Jin et al.'s approach.

We first analyze the Hessian operator in the high-frequency approximation in the case of single-fold surveys and in the case of triplicated ray fields. Then, we present the imaging condition under which such migrations produce artifact-free images and develop in this case migration/inversion formulas. We discuss the specific cases of common-shot or common-offset migration and show that the imaging condition does not hold generally in case of triplicated ray fields.

Asymptotic analysis of the Hessian

Consider the Hessian

$$\mathcal{B}^\dagger \mathcal{Q} \mathcal{B}(\mathbf{x}, \mathbf{x}_0) = \int_{\gamma_0} d\sigma \int d\omega \mathcal{B}^\dagger(\mathbf{r}(\sigma), \omega, \mathbf{s}(\sigma), \mathbf{x}) \times \mathcal{Q}(\mathbf{r}(\sigma), \omega, \mathbf{s}(\sigma), \mathbf{x}_0) \quad (9)$$

$$= \int d\omega \int_{\gamma_0} d\sigma \sum_{n=1}^{N(\sigma, \mathbf{x})} \sum_{n'=1}^{N'(\sigma, \mathbf{x}_0)} \mathcal{D}_n^{n'}(\sigma, \omega, \mathbf{x}, \mathbf{x}_0), \quad (10)$$

with

$$\mathcal{D}_n^{n'}(\sigma, \omega, \mathbf{x}, \mathbf{x}_0) = \mathcal{Q} |\mathcal{K}(\omega)|^2 \mathcal{U}_n^{n'}(\sigma, \mathbf{x}, \mathbf{x}_0) e^{i\omega \mathcal{F}_n^{n'}(\sigma, \mathbf{x}, \mathbf{x}_0)}, \quad (11)$$

with

$$\begin{cases} \mathcal{U}_n^{n'}(\sigma, \mathbf{x}, \mathbf{x}_0) = \mathcal{A}_n(\sigma, \mathbf{x}_0) \mathcal{A}_{n'}(\sigma, \mathbf{x}) \\ \mathcal{F}_n^{n'}(\sigma, \mathbf{x}, \mathbf{x}_0) = \mathcal{T}_{n'}(\sigma, \mathbf{x}_0) - \mathcal{T}_n(\sigma, \mathbf{x}) \end{cases}, \quad (12)$$

where $\mathcal{A}(\sigma, \mathbf{x}_0) = \mathcal{A}(\mathbf{r}(\sigma), \mathbf{x}_0, \mathbf{s}(\sigma))$, and $\mathcal{T}(\sigma, \mathbf{x}_0) = \mathcal{T}(\mathbf{r}(\sigma), \mathbf{x}_0, \mathbf{s}(\sigma))$.

The Hessian operator can be decomposed into singular and regular contributions (ten Kroode et al., 1994). Since we introduced high frequency asymptotic approximations for the estimation of the forward operator, we are only interested in the singular components of the Hessian.

We have now to carefully analyze the asymptotic behavior of the Hessian operator. In the linearized seismic inverse problem, Lailly (1984) and Tarantola (1984) showed that a migration is equivalent to the calculation of the gradient at the first iteration of a local optimization. Generally, the gradient is precisely estimated while the inverse Hessian, $\mathcal{H}^{-1}$, is only approximated by a diagonal operator, $\mathcal{H}_{est}^{-1}$ (Beydoun and Mendes, 1989; Jin et al., 1992; Chavent and Plessix, 1999). In this case, operator $\mathcal{H} \mathcal{H}_{est}^{-1}$ is the resolution operator associated with a single migration step (Operto et al., 1998). Since estimation of the inverse Hessian are diagonal operators, focused images can only be obtained if all the singular contributions of the

Hessian are located over the diagonal $\mathbf{x} = \mathbf{x}_0$ for identical ray trajectories, i.e., $n' = n$ (ten Kroode et al., 1994).

Ten Kroode et al. (1994) provided a very interesting and powerful analysis of the problem. In the case of multifold data, they demonstrated that the most singular contributions of the Hessian operator were localized along the diagonal $\mathbf{x} = \mathbf{x}_0$ for identical ray trajectories, i.e., $n' = n$. However, they also demonstrated that other singular contributions to the Hessian operator may also exist outside of the diagonal $\mathbf{x} \neq \mathbf{x}_0$ or for nonidentical ray trajectories, i.e., $n' \neq n$. Those singular contributions disappear under the traveltime injectivity condition (TIC). The analysis by ten Kroode et al. (1994) was limited to multifold data. A demonstration of ray + Born inversion on the multifold Marmousi data set was presented by Operto et al. (2000). The case of general acquisition geometries such as single-fold trace gathers was investigated by Nolan and Symes (1996) and Nolan (1996). A more general condition was established and renamed the imaging condition (Nolan and Symes, 1996).

Because the works of ten Kroode et al. (1994), Nolan and Symes (1996), and Nolan (1996) are quite difficult for geophysicists not familiar with the theory of Fourier integral operators and pseudodifferential operators (Treves, 1980), we try in the following section to express the imaging condition in a more physical way.

Imaging condition

Consider a single fold gather ($\gamma = \gamma_0$) parameterized by σ . Suppose that locally coherent events (Billette and Lambaré, 1998) can be identified in the gather. Such events are defined by a given trace position σ [$\mathbf{s}(\sigma)$ and $\mathbf{r}(\sigma)$], a two-way traveltime T , and the slope $p^\sigma = \partial T / \partial \sigma$ (Figures 2, 3).

The TIC, established by ten Kroode et al. (1994) for multifold gather, can be adapted to the case of a single-fold gather. It states that given $(\sigma, \gamma_0, T, p^\sigma)$, it is possible to uniquely reconstruct a couple of ray segments or, in other words, to uniquely define a scattering point $\mathbf{x}$ and the take-off angles β_s and β_r (Figure 2), or equivalently dip angle $\zeta = (\beta_s + \beta_r)/2$ and aperture angle $\theta = \beta_s - \beta_r$.

If the TIC is satisfied, all the locally coherent events in the data are focused at a single position after migration/inversion.

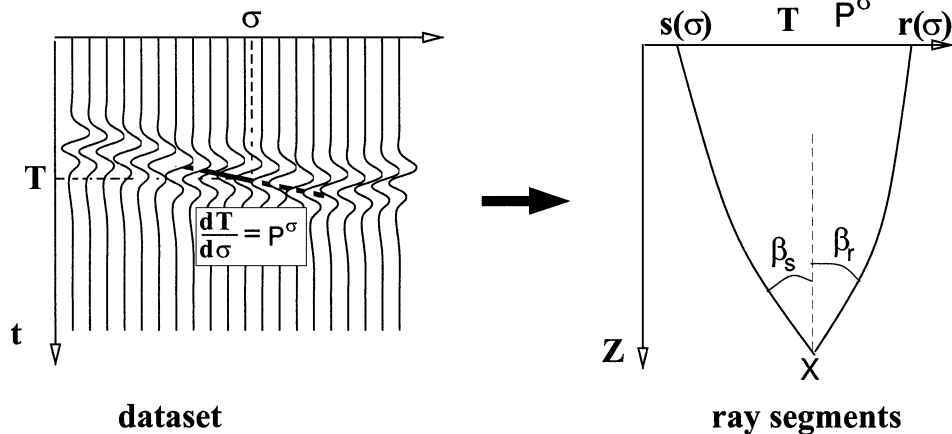


FIG. 2. Imaging condition for a single-fold dataset. The slope at the surface $\partial T / \partial \sigma(\sigma) = p^\sigma$, the shot and receiver positions $\mathbf{s}(\sigma)$ and $\mathbf{r}(\sigma)$, and the two-way traveltime $T(\sigma)$ define uniquely a reflected/scattered event given by a couple of ray segments.

In a later study, Nolan and Symes (1996) reconsidered the TIC in the context of general acquisition geometries. They derived an imaging condition that required in addition to the TIC a local differential expression of the TIC,

$$\det \left| \frac{\partial(\sigma, \gamma, T, p^\sigma)}{\partial(\mathbf{x}, \zeta, \theta)} \right| \neq 0, \quad (13)$$

for any $\mathbf{x}$, ζ and θ .

In case of multifold gather, the definition of locally coherent events requires an additional slope ($\gamma, \sigma, T, p^\sigma, p^\gamma$). This additional slope, $p^\gamma = \partial T / \partial \gamma$, allows us to constrain in a much better way the couple of ray segments [even more it provides information on the quality of the velocity macromodel (Billette and Lambaré, 1998)] and the imaging condition generally holds even for triplicated ray fields, except for some tortuous ray trajectories (ten Kroode et al., 1994).

In the case of a single-fold gather, where a single slope is available in the data set, the imaging condition may break down for triplicated ray fields much more frequently than for multifold data. As a result, artifacts in the migrated images may happen (Nolan, 1996; Nolan and Symes, 1996; Xu et al., 1998), and later examples will show both kinematic and amplitude effects.

Migration/inversion formulas

To derive migration/inversion formulas, let us suppose that the imaging condition is satisfied. It ensures that the singular part of the Hessian operator [equation 10] can be obtained with a local analysis around the diagonal terms $\mathbf{x} = \mathbf{x}_0$ and for $n' = n$.

The Hessian operator can be approximated up to smooth operators by

$$B^\dagger \mathcal{Q} B(\mathbf{x}, \mathbf{x}_0) \approx \int d\omega \int_{\gamma_0} d\sigma \sum_{n=1}^{N(\sigma, x_0)} \mathcal{Q} |\mathcal{K}(\omega)|^2 \mathcal{A}_n^2(\sigma, x_0) e^{-i\omega \nabla T_n(\sigma, x_0) \cdot (\mathbf{x} - \mathbf{x}_0)}, \quad (14)$$

where we used local approximations for the amplitude and phase terms in equation (12) (Miller et al., 1987; Jin et al., 1992)

$$\begin{cases} \mathcal{U}_n^n(\sigma, \mathbf{x}, \mathbf{x}_0) \approx \mathcal{A}_n^2(\sigma, \mathbf{x}_0) \\ \mathcal{F}_n^n(\sigma, \mathbf{x}, \mathbf{x}_0) \approx -\nabla T_n(\sigma, \mathbf{x}) \cdot (\mathbf{x} - \mathbf{x}_0) \end{cases} \quad (15)$$

The integral operator (14) includes a kernel which reminds us the integral expression of the 2-D Dirac function,

$$\delta(\mathbf{x} - \mathbf{x}_0) = \frac{1}{(2\pi)^2} \iint_{\mathbb{R}^2} d\mathbf{k} e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}_0)}. \quad (16)$$

The analogy between the asymptotic form of the Hessian (14) and the Dirac delta function (16) can be used to obtain an analytic diagonal expression of the inverse Hessian (Jin et al., 1992; Thierry et al., 1999b). We choose for the $\mathcal{Q}$ factor

$$\mathcal{Q}(\sigma, \mathbf{x}_0, \omega) = \frac{1}{(2\pi)^2} \frac{1}{|\mathcal{K}|^2} \left| \frac{\partial \zeta}{\partial \sigma} \right|_n \left| \frac{\partial(\mathbf{k})}{\partial(\zeta, \omega)} \right|, \quad (17)$$

with $\mathbf{k} = \omega \mathbf{q}$. At this point, we could use the Beylkin's determinant $\mathfrak{B}_n = \det(\mathbf{q}, \partial \mathbf{q} / \partial \sigma)$ (Bleistein, 1987) using the decomposition

$$\left| \frac{\partial \zeta}{\partial \sigma} \right|_n \left| \frac{\partial(\mathbf{k})}{\partial(\zeta, \omega)} \right| = \left| \frac{\partial(\mathbf{k})}{\partial(\sigma, \omega)} \right|_n = |\omega| |\mathfrak{B}_n|, \quad (18)$$

but for later simplifications we prefer the decomposition given in expression (17). The approximate Hessian becomes

$$B^\dagger \mathcal{Q} B(\mathbf{x}, \mathbf{x}_0) \approx \frac{1}{(2\pi)^2} \int d\omega \int d\sigma \sum_{n=1}^{N(\sigma, x_0)} \left| \frac{\partial \zeta}{\partial \sigma} \right|_n \left| \frac{\partial(\mathbf{k})}{\partial(\zeta, \omega)} \right| e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}_0)} \quad (19)$$

$$= \frac{1}{(2\pi)^2} \int d\omega \sum_{n=1}^{N(\sigma, x_0)} \int_{\zeta_{\min}^n}^{\zeta_{\max}^n} d\zeta \left| \frac{\partial(\mathbf{k})}{\partial(\zeta, \omega)} \right| e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}_0)}. \quad (20)$$

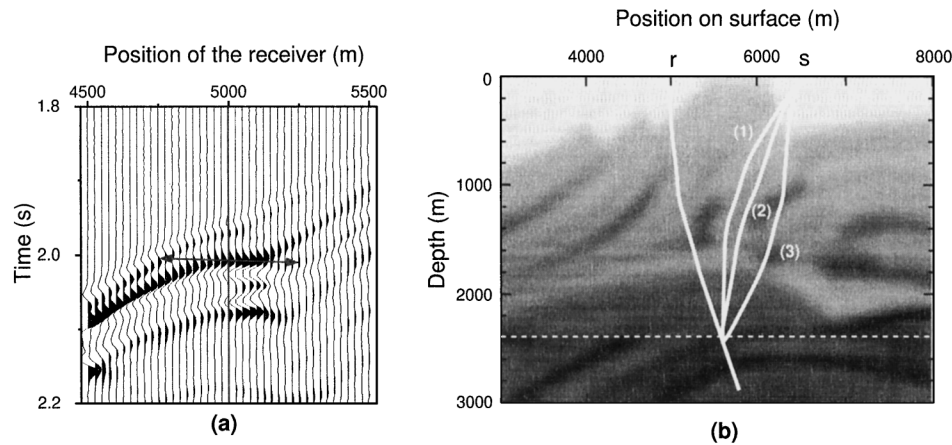


FIG. 3. Imaging condition for a common shot. (a) The shot gather allows to determine, for a given locally coherent event, the double traveltimes, the positions of the source and the receiver, and the slope at the receiver position. (b) However, different ray segments denoted by (1), (2), and (3) exist which could perfectly explain the data. The three diffracting points do not exactly have the same positions. The velocity model and synthetic data set are the same as used below.

Due to the imaging condition, it can be demonstrated (see appendix B) that under the condition of no grazing nor direct rays, we have $\frac{\partial \zeta}{\partial \sigma}|_{\mathbf{x}_Y} \neq 0$. As a consequence, contributions $[\zeta_{\min}^n, \zeta_{\max}^n]$ do not overlap but complement (the direct consequence of this status is that contributions of the various branches have to be added to the image rather than averaged!), and we can do the approximation

$$B^\dagger \mathcal{Q}B(\mathbf{x}, \mathbf{x}_0) \approx \frac{1}{(2\pi)^2} \int d\omega \int_{\zeta_{\min}}^{\zeta_{\max}} d\zeta \left| \frac{\partial(\mathbf{k})}{\partial(\zeta, \omega)} \right| e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}_0)} \quad (21)$$

$$\approx \frac{1}{(2\pi)^2} \iint d\mathbf{k} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{x}_0)} = \delta(\mathbf{x} - \mathbf{x}_0), \quad (22)$$

with $\zeta_{\max}$ and $\zeta_{\min}$ being the maximum and minimum of the n branches boundaries in dip. The final ray + Born migration/inversion formula is finally given by the gradient

$$\delta m|_{\gamma_0}(\mathbf{x}) \approx \int_{\gamma_0} d\sigma \sum_{n=1}^{N(\sigma, \mathbf{x})} \mathcal{E}_n^{\gamma_0}(\sigma, \mathbf{x}) \text{Hilb}^{(1-\alpha_n(\sigma, \mathbf{x}))} [\delta \mathcal{G}_{\text{obs}}](\sigma, T_n(\sigma, \mathbf{x})), \quad (23)$$

where Hilb^m denotes the Hilbert transform to the order m , and the amplitude of the quantitative migration operator is defined by

$$\mathcal{E}_n^{\gamma_0}(\sigma, \mathbf{x}) = \frac{|\mathbf{q}|_n^2 \left| \frac{\partial \zeta}{\partial \sigma} \right|_n}{2\pi \mathcal{A}_n(\mathbf{r}, \mathbf{x}, \mathbf{s})}. \quad (24)$$

We used

$$\left| \frac{\partial(\mathbf{k})}{\partial(\zeta, \omega)} \right| = |\omega| |\mathbf{q}|^2. \quad (25)$$

A ray + Kirchhoff migration/inversion formula can be derived similarly for the specular reflectivity function

$$R|_{\gamma_0}(\mathbf{x}) \approx \int_{\gamma_0} d\sigma \sum_{n=1}^{N(\sigma, \mathbf{x})} \frac{\mathcal{E}_n^{\gamma_0}(\sigma, \mathbf{x})}{|\mathbf{q}|_n} \text{Hilb}^{(1-\alpha_n(\sigma, \mathbf{x}))} \left[\frac{\partial \delta \mathcal{G}_{\text{obs}}}{\partial t} \right](\sigma, T_n(\sigma, \mathbf{x})). \quad (26)$$

Several choices can be given for σ depending on the choice of the single-fold dataset.

For a common-shot gather, we have $\gamma_0 = \mathbf{s}_0$ and $\sigma = \mathbf{r}$ (the receiver position). The weight of the migration/inversion kernel is thus

$$\mathcal{E}_n^{\mathbf{s}_0}(\mathbf{r}, \mathbf{x}) = \frac{|\mathbf{q}|_n^2 \left| \frac{\partial \beta_{\mathbf{r}}}{\partial \mathbf{r}} \right|_n}{4\pi \mathcal{A}_n(\mathbf{r}, \mathbf{x}, \mathbf{s})}, \quad (27)$$

which can be given by paraxial ray tracing (Farra and Madariaga, 1987; Lambaré et al., 1996). It corresponds to the migration/inversion formula of Bleistein (1987) and Miller et al. (1987).

For a common-offset gather, we have $\gamma_0 = h_0 = (\mathbf{s} - \mathbf{r})$ and $\sigma = \mathbf{X} = (\mathbf{s} + \mathbf{r})/2$ (the common-midpoint position). We have

for the weight of the migration/inversion kernel

$$\mathcal{E}_n^{\mathbf{h}_0}(\mathbf{X}, \mathbf{x}) = \frac{|\mathbf{q}|_n^2 \left| \frac{\partial \beta_{\mathbf{s}}}{\partial \mathbf{s}} + \frac{\partial \beta_{\mathbf{r}}}{\partial \mathbf{r}} \right|_n}{4\pi \mathcal{A}_n(\mathbf{r}, \mathbf{x}, \mathbf{s})}, \quad (28)$$

which again corresponds to the migration/inversion formula of Bleistein (1987) and Miller et al. (1987).

Failure of the imaging condition

We discuss now the conditions for a violation of the imaging condition in cases of common-shot and common-offset migrations (Nolan, 1996; Nolan and Symes, 1996). For that discussion, the use of the local differential version of the TIC is very helpful. In appendix A, we demonstrate that condition (13), for any $\mathbf{x}$, ζ , and, θ , is equivalent to

$$\mathbf{q}^2 \frac{\partial \gamma}{\partial \theta} \Big|_{\mathbf{x}, \zeta} \neq 0. \quad (29)$$

In the common-shot case, the imaging condition will be infringed for direct rays, $\mathbf{q}^2 = 0$, but also if

$$\frac{\partial \mathbf{s}}{\partial \theta} \Big|_{\mathbf{x}, \zeta} = \frac{1}{2} \frac{\partial \mathbf{s}}{\partial \beta_{\mathbf{s}}} \Big|_{\mathbf{x}} = 0. \quad (30)$$

Figure 4 illustrates the fact that in case of caustics in the ray field from the shot point $\mathbf{s}$, there may be an ambiguity about the localization of the couple of ray segments $(\mathbf{x}, \beta_{\mathbf{s}}, \beta_{\mathbf{r}})$ associated with a locally coherent event $(\mathbf{s}, \mathbf{r}, \mathbf{p}^r, T)$.

In case of a common-offset gather, a similar analysis can be developed. The local slope is the horizontal component at the surface of the midpoint slowness vector, $(\mathbf{p}^s + \mathbf{p}^r)/2$ (Figure 4). The imaging condition will be infringed for direct rays, $\mathbf{q}^2 = 0$, but also if

$$\frac{\partial \mathbf{h}}{\partial \theta} \Big|_{\mathbf{x}, \zeta} = \frac{\partial \mathbf{s}}{\partial \beta_{\mathbf{s}}} \Big|_{\mathbf{x}} + \frac{\partial \mathbf{r}}{\partial \beta_{\mathbf{r}}} \Big|_{\mathbf{x}} = 0. \quad (31)$$

The determination of the cases where the imaging condition fails is not so simple as for the common shot case. As we will see in the following examples, common-offset imaging generally fails in case of triplicated ray fields.

When the imaging condition fails, two kinds of artifacts may appear in the migrated images:

- 1) Singular nondiagonal events appear in the Hessian. They are less singular than the diagonal terms, but still produce

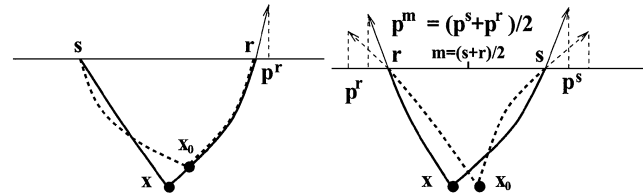


FIG. 4. Failure of the imaging condition. Left: in common shot imaging, a locally coherent event can be interpreted by two pairs of ray segments, reaching the surface with the same source and receiver positions, the same double traveltimes and the same slope at the receiver position. Right: n common-offset imaging, a locally coherent event can be interpreted by two pairs of ray segments; $\mathbf{m}$ denotes the midpoint.

focusing of events at a wrong position. Locally coherent events in the constant γ_0 gather may not be unambiguously associated with a single pair of ray segments.

- 2) Concerning the diagonal terms of the Hessian, the migration/inversion formulas were established with the hypothesis that $\partial\zeta/\partial\sigma|_{x,y} \neq 0$ [to derive equation (22) from equation (21)]. When the imaging condition is not satisfied, we can not be sure anymore if the various contributions $[\zeta_{\min}^n, \zeta_{\max}^n]$ overlap or not, and the amplitude of the kernel of the migration/inversion formula may be totally erroneous.

CANONICAL TEST CASE

In order to illustrate these phenomena, a canonical test was built. Data were obtained by the difference of two data sets generated on two velocity models (both constant density acoustic). The first model is a smoothed version of the Marmousi model (Bourgeois et al., 1991) low-pass filtered to 150 m, i.e.,

the velocity macromodel required for the migration/inversion of the Marmousi dataset (Operto et al., 2000). The second model is the same smoothed model where a 10 m/s perturbation layer has been superimposed between depths 2400 and 2500 m (Figure 5). Thus, the difference of the two data sets will allow us to test the migration of the layer in the deeper part with a complex overburden.

The two velocity models were discretized to 5 m in both x and z for forward modeling with finite differences of the wave equation (Noble, 1992). The acquisition geometry was almost the same as for the Marmousi experiment (Bourgeois et al., 1991), with only three slight differences: first offset is at 100 m (instead of 200 m), and source and receiver positions are at 25 m in depth and with a 56 ms source delay. The bandwidth of the source is [5, 13, 40, 55] Hz. A common-shot gather and a common-offset gather are displayed on Figure 6.

Common-shot and common-offset ray + Born migration/inversions were applied to image this layer. The background model was used as velocity macromodel. Many triplications

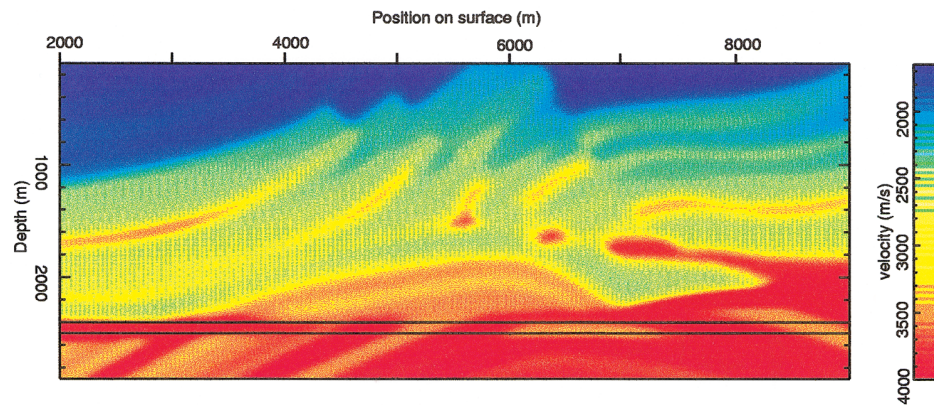


FIG. 5. Canonical model, consisting of two velocity models. The first model is the Marmousi model (Bourgeois et al., 1991) low-pass filtered to 150 m [velocity macromodel required for migration/inversion of the Marmousi dataset (Operto et al., 2000)]. The second model is the same velocity model with a superimposed 10 m/s perturbation layer that we want to image. The position of the perturbation, between depths 2400 and 2500 m, is indicated by two horizontal solid lines.

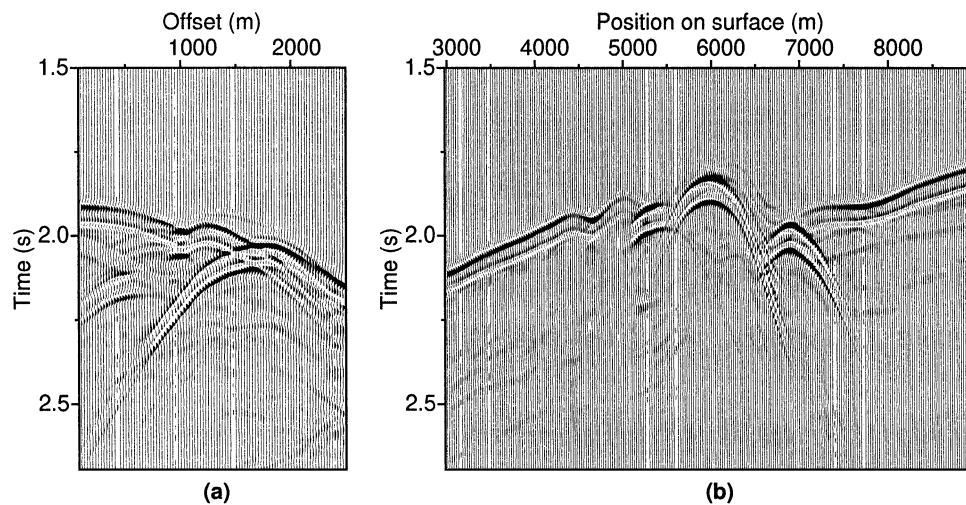


FIG. 6. Canonical model with (a) common-shot (for source at position 7500 m) and (b) common-offset (for the first offset 100 m) gathers. As these data were obtained by the difference of two data sets modeled in a complex medium (see the text for more details), no information is available before 1.8 s.

arise in the complex part of the model (Figure 7). Migration parameter maps (traveltime, amplitude, angle, slowness vector, KMAH index, paraxial derivatives, etc.) were computed using a wavefront construction code (Lambaré et al., 1996). A uniform ray-density criterion insures the accuracy of the multivalued maps even in the triplicated zones. Migration/inversion codes were not designed for CPU efficiency, but simply to demonstrate the advantages of common-diffracting-angle imaging. Embedded interpolations, as they were introduced by Thierry et al. (1999b), are not used and, indeed, they would be difficult to implement due to multivaluedness. No anti-alias filter was applied and, in addition to the elimination of up-going rays, the only specificity of our migration/inversion code is a limitation of the trace contributions to a 5000-m zone around the midpoint position.

We do ray + Born migration/inversion, and each depth profile represents the short wavelength components of the velocity perturbations. Except when it is explicitly mentioned, all the panels are plotted with the same clip (about one half of the exact velocity perturbation) in order to emphasize the quantitative aspects of migration/inversion.

CIGs in shot and offset for $X = 3500$ m (a priori in the noncomplex part of the model) are displayed on Figure 8. Up to $X = 5000$ m, CIGs calculated in the shot domain are of good quality. Beyond $X = 5000$ m, they exhibit strong artifacts as it can be observed everywhere on the CIG in offset. In fact, as it appears on Figure 7, position $X = 3500$ m, which could be thought to be in a calm zone of the model, is affected by triplications coming from guided propagations in ray shooting position, beyond $X = 5000$ m. The CIG in offset that take

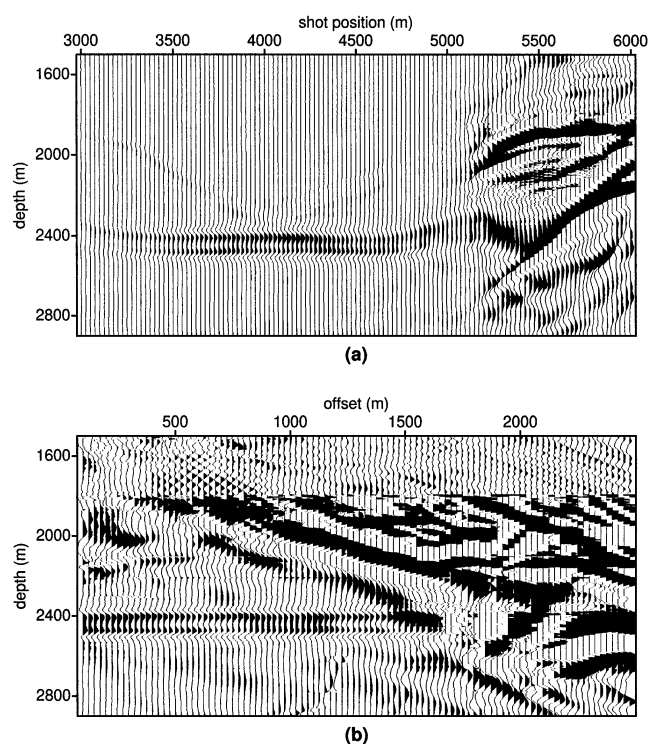


FIG. 8. CIGs in shot domain (a) and offset domain (b) for the a priori noncomplex zone ($X = 3500$ m) of the canonical model. The CIGs are plotted with a clip fixed to half of the velocity perturbation in the layer.

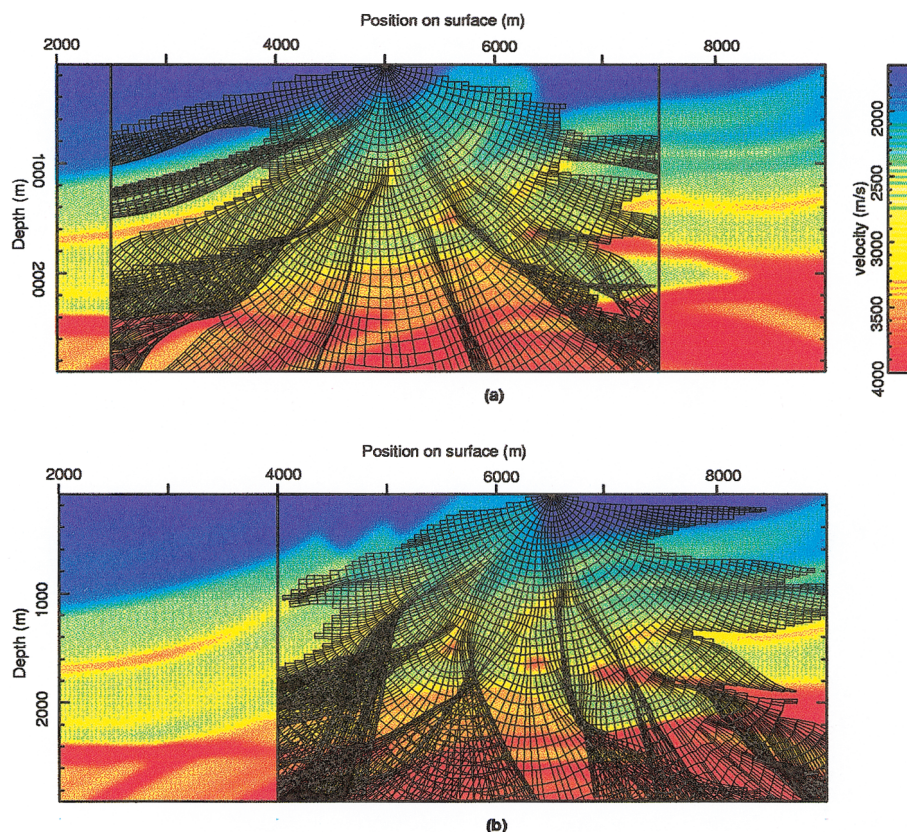


FIG. 7. Two wavefront constructions in the smooth Marmousi model, for a source at positions 5000 m (a) and 6500 m (b). Many triplications appear due to the complexity of the velocity model.

into account these contributions for each offset class is then strongly altered even if they do not correspond to specular contributions. When limiting the contributing shots to the first 70 (which is largely sufficient in terms of specular contributions), the resulting CIG in offset is good. Figure 9 shows these CIGs in shot and offset at position $X = 3500$ m illustrating the well-known advantage of CIG in offset in terms of boundary effects (Ehinger et al., 1996).

CIGs in shot and offset for $X = 6200$ m (in the complex part of the model) are displayed on Figure 10. Both are affected by strong artifacts. For the CIG in the shot domain, the image of the layer is altered by a strong convex shape, as around position 7400 m. This corresponds to a kinematical artifact resulting from the singular nondiagonal terms of the Hessian (images are not well focused!). For the CIG in the offset domain, the image is completely oversaturated (amplitudes overestimated). It is a dynamical artifact resulting from the erroneous estimation of the diagonal term of the Hessian. Strong convex shapes (kinematic artifacts) appear now on both CIGs.

Finally, Figures 11 and 12 show various individual common-shot image panels and common-offset image panels over the full target. Strong artifacts appear, kinematic artifacts are also revealed for the common-offset case when results are clipped with respect to the maximum of the plot (percentile 99), as in Figure 13.

This canonical example illustrates both kinds of artifact when doing common-shot or common-offset migration/inversion. It appears that common-offset migration/inversion is particularly unstable in the presence of triplications.

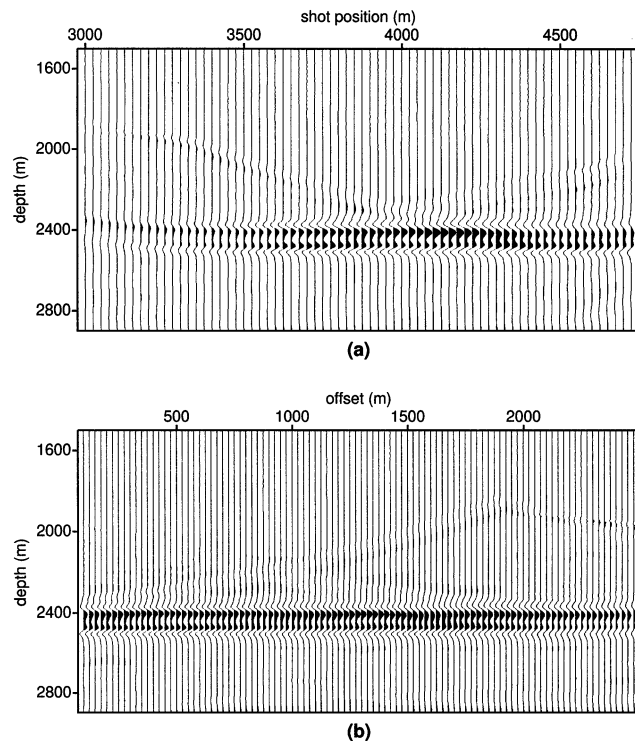


FIG. 9. CIGs in shot domain (a) and offset domain (b) in the noncomplex zone ($X = 3500$ m) of the canonical model, using only the 70 first shots during the migration. The CIGs are plotted with the a clip fixed to half of the velocity perturbation in the layer.

COMMON-DIFFRACTING-ANGLE MIGRATION/INVERSION

A solution for migration in complex media

From the previous study, it is clear that in case of triplications, we can not be sure to obtain clean images when doing common-shot or common-offset migration/inversion. The problem is to obtain clean CIGs for AVO or velocity analysis.

A solution has been proposed for migration/inversion in anisotropic media where caustics may develop: common-diffracting-angle (θ) migration/inversion (de Hoop et al., 1994). The relation between the couple of dip and aperture angles (ζ, θ) and a trace position at surface (s, r) depends on the reflection/diffraction point $\mathbf{x}_0$ (Figure 1), and consequently depends on each point in the image. The decomposition in common-angle sections is not done a priori on the multifold data set but in the depth domain inside the migration loop. The summation is done over the dip angle $\sigma = \zeta$ and the sorting for every diffracting angle $\gamma = \theta$.

In the context of single-fold migration/inversion in complex media, common diffracting angle migration seems very interesting. The imaging condition therefore holds, whatever the presence of caustics:

- 1) For any point $\mathbf{x}_0$ in the image, the pair of ray segments is completely and uniquely defined by the two angles (θ, ζ) (except in case of grazing angles at the surface where the length of the rays segments has to be specified);
- 2) Local differential version of the TIC (29) is always valid (except for direct rays).

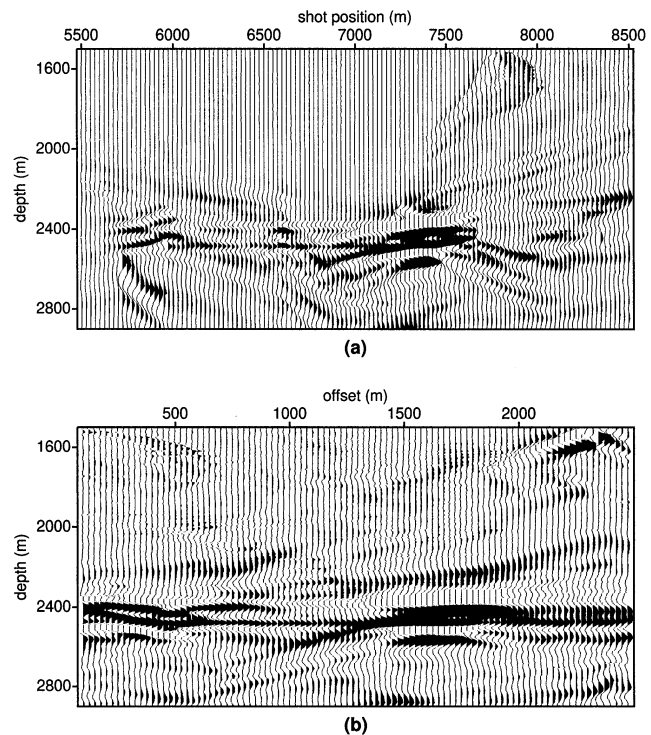


FIG. 10. CIGs in shot domain (a) and offset domain (b) in the complex part ($X = 6200$ m) of the canonical model. The CIGs are plotted with a clip fixed to half of the velocity perturbation in the layer.

Let us now draw a curious conclusion. In a single-fold sub-data set, local differential condition (29) implies that any zero offset trace corresponding to ray shooting positions with triplications will violate the imaging condition. It means that in a complex model, zero-offset gathers can not provide artifact-free depth images. On the contrary, there is no problem with $\theta = 0$ imaging that uses the same traces. This apparent contradiction comes from the fact that in common-zero-offset migration, all the cross contributions between branches reaching the surface with different angles are taken into account, whereas in common-zero-angle migration, rays coming to the source and receiver points are identical.

Common-diffracting-angle migration/inversion formula

The common-diffracting-angle migration/inversion formula can be derived easily from the former developments. We use local considerations around a point $\mathbf{x}_0$, and we simply have to use $\gamma_0 = \theta_0$ and $\sigma = \zeta$ (necessary condition $\frac{\partial \zeta}{\partial \sigma}|_{\mathbf{x}_0} \neq 0$ is obviously satisfied) in formulas (23) and (26). We have for the weight of the migration/inversion kernel (24)

$$\mathcal{E}_n^{\theta_0}(\zeta, \mathbf{x}) = \frac{|\mathbf{q}|_n^2}{2\pi \mathcal{A}_n(\mathbf{r}, \mathbf{x}, \mathbf{s})}. \quad (32)$$

Imaging with common-diffracting angle is however not equivalent practically to common-shot or common-offset imaging. As we mentioned before, data can not be sorted a priori in a common-angle data set. The sorting is done inside the migration/inversion loop for each position, $\mathbf{x}$, in the target, and all traces have to be considered a priori. Rather than migration formula (23) with a summation over ζ , we have to use a migration formula over $\mathbf{s}$ and $\mathbf{r}$ with an extraction of the considered θ_0 panel. We get for the ray + Born common-angle migration/inversion formula

$$\delta m|_{\theta_0}(\mathbf{x}) \approx \int_{\theta_0} d\zeta \sum_{n=1}^N \mathcal{E}_n^{\theta_0} \text{Hilb}^{(1-\alpha_n)} [\delta \mathcal{G}_{\text{obs}}](T_n) \quad (33)$$

$$= \iint d\mathbf{s} d\mathbf{r} \sum_{n=1}^N \delta(\theta - \theta_0) \mathcal{E}'_n \text{Hilb}^{(1-\alpha_n)} [\delta \mathcal{G}_{\text{obs}}](T_n), \quad (34)$$

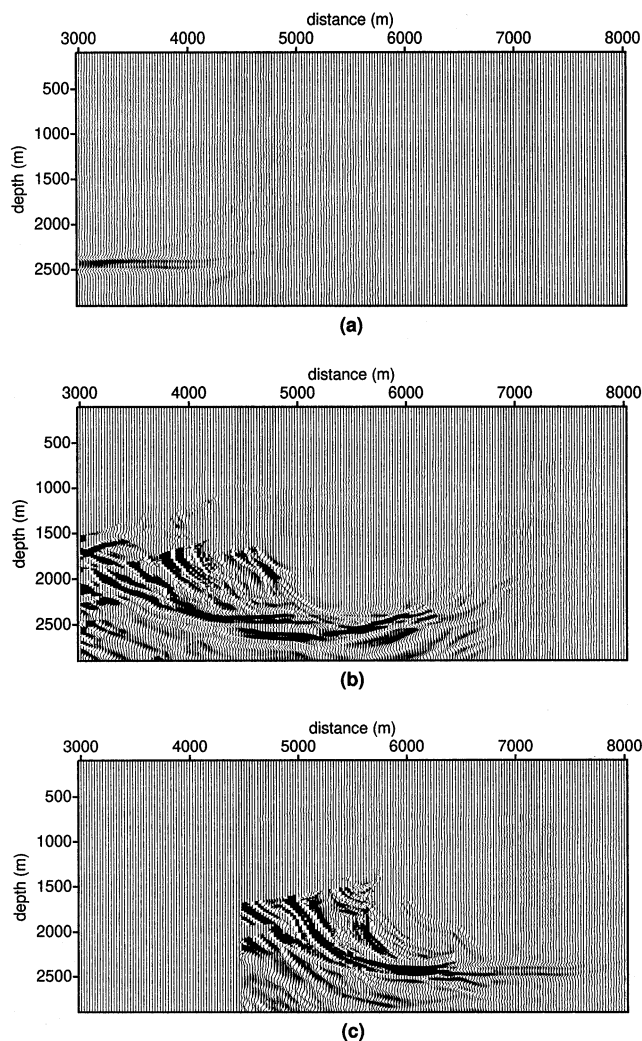


FIG. 11. Common-shot panels for shots $X = 4000$ m (a), $X = 6000$ m (b), and $X = 7500$ m (c). Profiles are plotted with a clip fixed to half of the velocity perturbation in the layer.

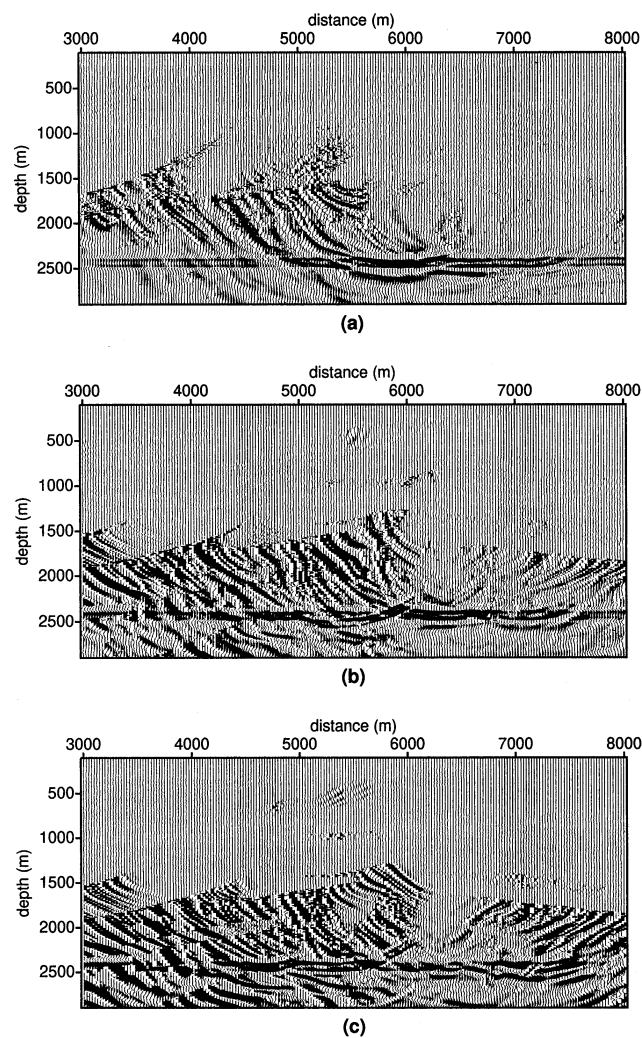


FIG. 12. Common-offset panels for offsets $h = 100$ m (a), $h = 850$ m (b), and $h = 2475$ m (c). Profiles are plotted with a clip fixed to half of the velocity perturbation in the layer.

and for the ray + Kirchhoff migration inversion formula

$$R|_{\theta_0}(\mathbf{x}) \approx \iint ds d\mathbf{r} \sum_{n=1}^N \delta(\theta - \theta_0) \frac{\mathcal{E}'_n}{|\mathbf{q}|_n} \text{Hilb}^{(1-\alpha_n)} \left[\frac{\partial \delta \mathcal{G}_{\text{obs}}}{\partial t} \right] (T_n), \quad (35)$$

with

$$\mathcal{E}'_n = \left| \frac{\partial(\zeta, \theta)}{\partial(\mathbf{s}, \mathbf{r})} \right|_n \frac{|\mathbf{q}|_n^2}{2\pi \mathcal{A}_n}, \quad (36)$$

which can be used as the common-diffracting-angle migration/inversion formula in case of standard surface acquisition lines. The Jacobian $\left| \frac{\partial(\zeta, \theta)}{\partial(\mathbf{s}, \mathbf{r})} \right|$ can be estimated by paraxial ray tracing using decomposition

$$\left| \frac{\partial(\zeta, \theta)}{\partial(\mathbf{s}, \mathbf{r})} \right| = \left| \frac{\partial \beta_s}{\partial \mathbf{s}} \right| \left| \frac{\partial \beta_r}{\partial \mathbf{r}} \right|. \quad (37)$$

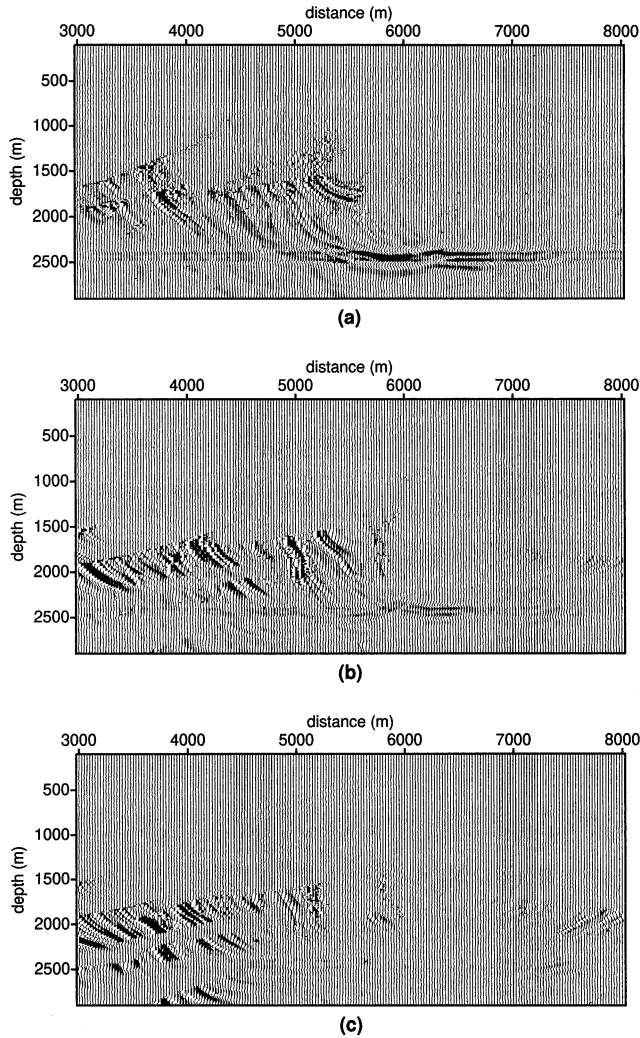


FIG. 13. Common-offset panels for offsets $h=100$ m (a), $h=850$ m (b), and $h=2475$ m (c) in the canonical example. Same results as Figure 13 clipped with respect to the maximum value of the plot (percentile 99), showing also kinematical artifacts.

Demonstration on the canonical test case

A common-diffracting-angle ray + Born migration/inversion code was developed. Continuous migration/inversion formula (34) has to be discretized. The Dirac function is simply replaced by a “door” function defined with the Heavyside function H as follows:

$$\delta(\theta - \theta_0) \approx \frac{H(\theta - \theta_0 + \Delta\theta/2)H(\theta_0 + \Delta\theta/2 - \theta)}{\Delta\theta}, \quad (38)$$

providing a normalized image for the angle class $[\theta_0 - \Delta\theta/2, \theta_0 + \Delta\theta/2]$. A smoother function could be used for avoiding aliasing effects due to the surface sampling.

CIGs in angle are displayed on Figure 14 for positions $X=3500$ m and $X=6200$ m, whereas Figure 15 shows various individual common-angle image panels over the full target. Results are incomparably better than those obtained with common-shot or common-offset migration/inversion. In the complex zone, $X=6200$ m, the CIG in angle is only altered by some boundary effects that may happen almost everywhere in the CIG due to the complexity of the ray field. The CIG in angle directly provides information for amplitude-versus-angle (AVA) analysis.

APPLICATION TO MARMOUSI

As a final demonstration, we propose an application to the Marmousi data set (Bourgeois et al., 1991). The Marmousi data set is generally considered as a reference test for imaging in complex media (Audebert et al., 1997). In terms of migration, very good results have been obtained by wave-equation

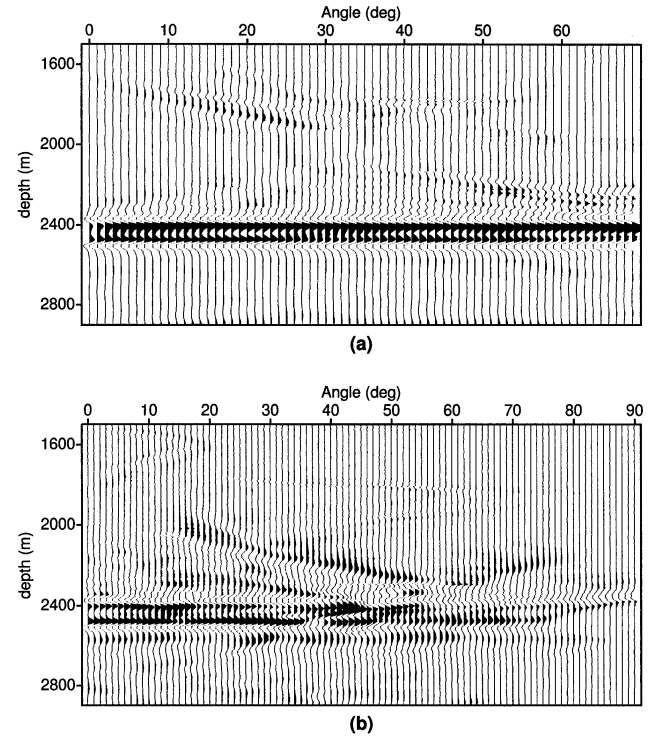


FIG. 14. CIGs in angle (a) in the noncomplex part ($X=3500$ m) and (b) in the complex part ($X=6200$ m) of the canonical model. CIGs are plotted with a clip fixed to half of the velocity perturbation in the layer.

migration (Versteeg, 1993). With Kirchhoff migration, results were obtained using kinematic information of the first arrival (Geoltrain and Brac, 1993). Image was poor in the complex deep part of the model. Results can be slightly improved by the use of the strongest arrival (Ettrich and Gajewski, 1996;

Thierry et al., 1999b) but for an optimal result the use of all the arrivals is unavoidable (Bevc, 1997; Operto et al., 2000). These results concerned migration of the full multifold data set. From the work of Duquet (1996), it was evident that CIGs obtained on the Marmousi data with wave-equation migration in the exact velocity model were not flat and altered by artifacts. No precise analysis of the phenomena was possible with wave-equation migration and, to remedy the situation, the authors proposed to add an extra condition in the migration kernel in order to enforce the flatness of the CIG.

For our application with the Marmousi data set, we use the same migration/inversion codes and the same velocity macro-model as for the canonical case. Data are deconvolved from source signature and water-bottom reverberations (Operto et al., 2000) (Figure 16).

CIGs in shot, offset, and diffracting angle are displayed in Figure 17 for positions $X = 3500$ m and $X = 6200$ m, leading to the same remarks as for the canonical case about the quality of the results. The target now covers the whole depth profile, and we notice (1) the variation of the angular aperture with depth, which increases first due to the mute and later decreases due to the limited offset; and (2) the brutal degradation of the CIG in offset, and to a lesser extent of the CIG in shot.

As soon as there are triplications, common-offset migration/inversion (Bleistein, 1987) dramatically fails. In Figure 17, the CIGs are plotted with the same clip enhancing the dynamical artifacts of migration/inversion. In Figure 18, the CIGs are plotted with clip fixed by percentile 98, showing kinematic artifacts. As for the canonical test, the selection of the first 70 shot gathers allows a far better image (Figure 19) as no triplication alters the image. Finally, Figures 20–23 show various common-shot, common-offset, and common-angle images over the full target. Strong artifacts appear in the common-shot and common-offset images. Far better results are obtained for the common-angle images. Residual artifacts, however, remain due to the boundaries of the acquisition geometry.

CONCLUSION

With two demonstrative synthetic examples, we show that both dynamic and kinematic artifacts appear in common-shot and common-offset migration/inversion in complex media. While dynamic artifacts (strong overestimation of migrated

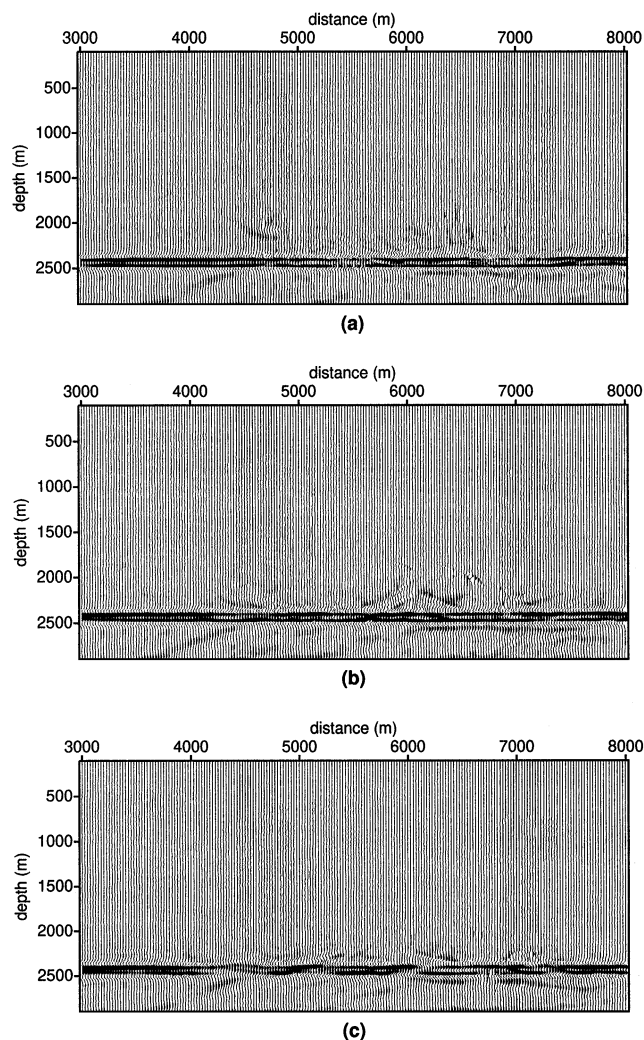


FIG. 15. Common-angle panels for angles $\theta = 15^\circ$ (a), $\theta = 45^\circ$ (b), and $\theta = 65^\circ$ (c). Clip is fixed to half of the velocity perturbation in the layer.

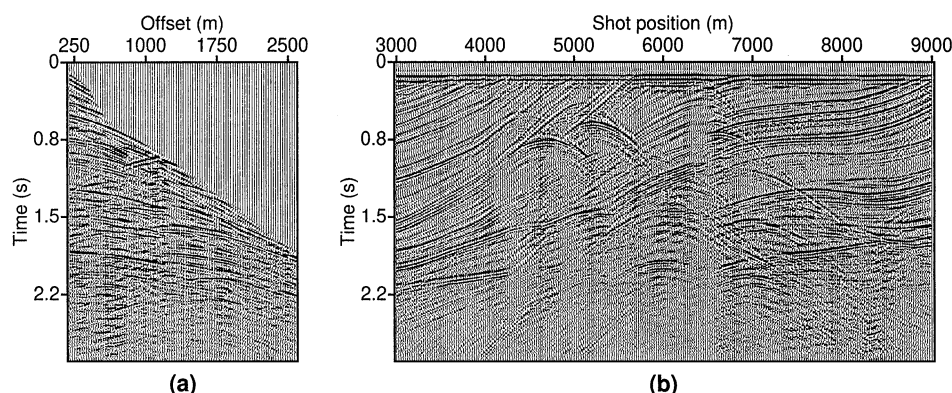


FIG. 16. Marmousi single-fold data sets. Common-shot gather (a) is represented for shot position at $X = 7500$ m. Common-offset gather (b) corresponds to the first offset $h = 200$ m.

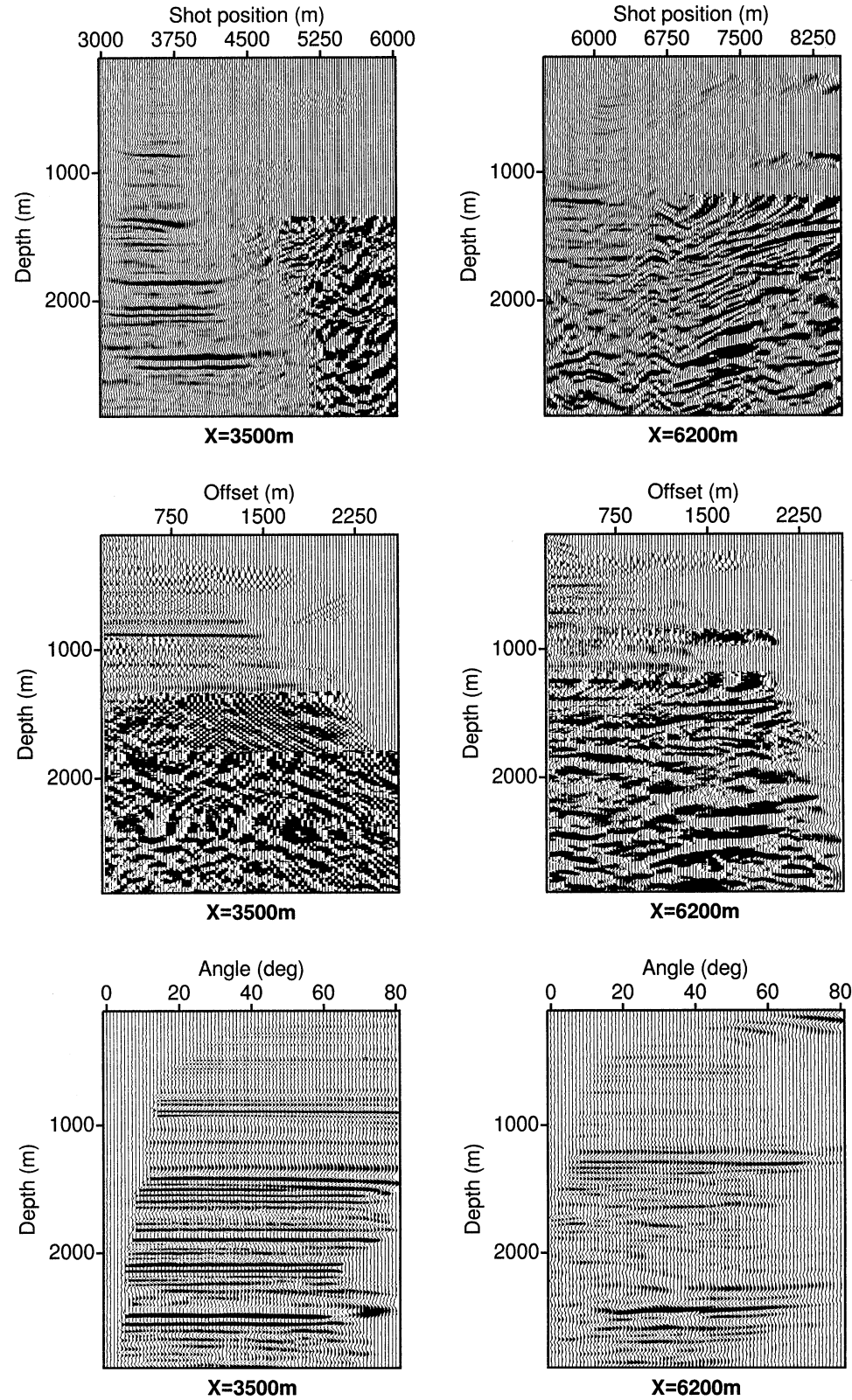


FIG. 17. CIGs in shot, offset, and angle in the noncomplex ($X = 3500$ m) part and in the complex ($X = 6200$ m) part of the Marmousi model. Results are displayed in relative impedance perturbation and are plotted with the same clip. Dynamic artifacts appear, except for the common-angle case where the imperfections in the image are due to the boundaries of the acquisition geometry at surface.

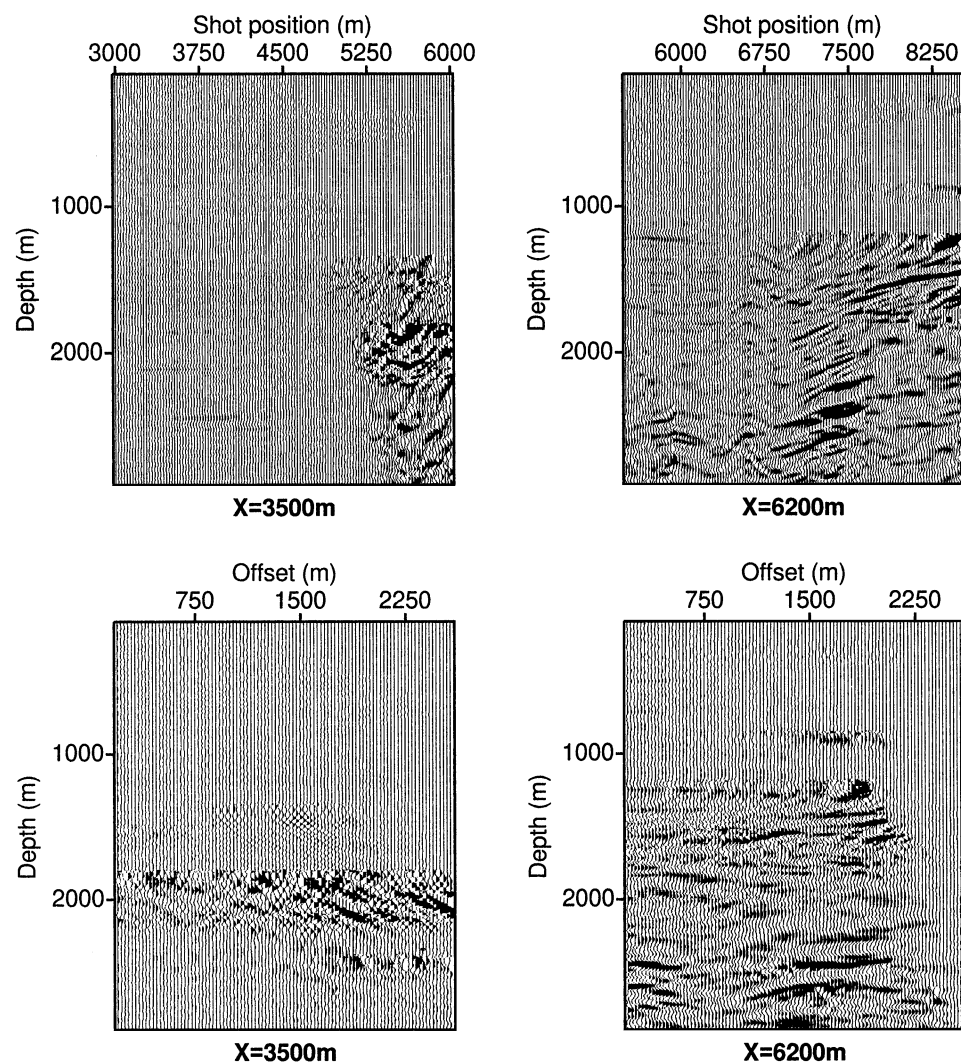


FIG. 18. CIGs in shot and offset in the noncomplex ($X = 3500\text{ m}$) part and in the complex ($X = 6200\text{ m}$) part of the Marmousi model. Results are displayed in relative impedance perturbation and are plotted with a clip fixed to percentile 98. Kinematic artifacts appear.

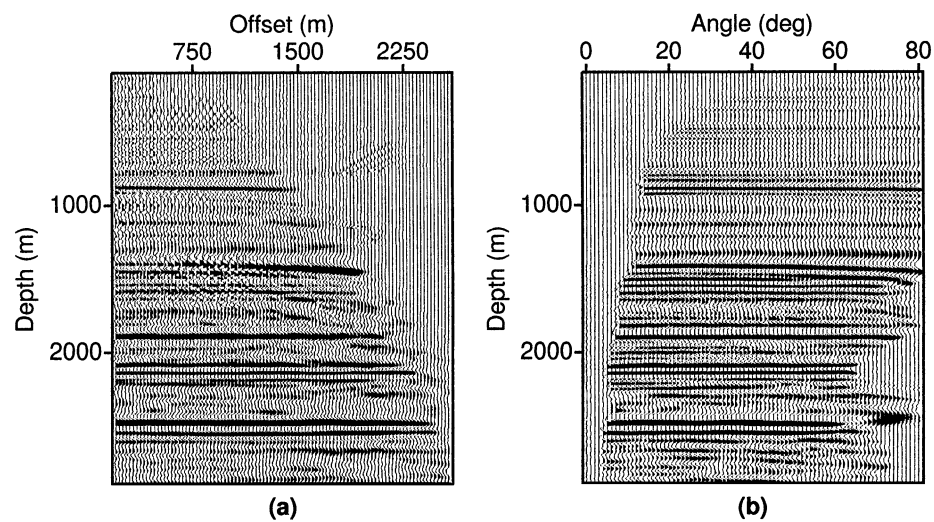


FIG. 19. CIGs in offset (a) and in angle (b) in the noncomplex ($X = 3500\text{ m}$) part of the Marmousi model. The CIG in offset was computed just using the first 70 shots for migration. Results are displayed in relative impedance perturbation and are plotted with the same clip.

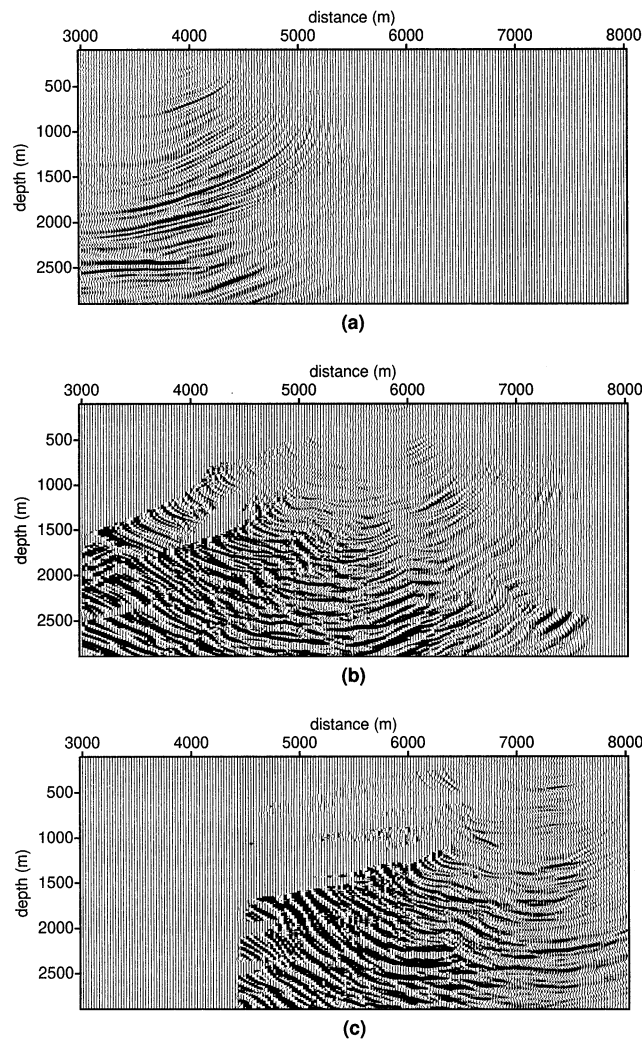


FIG. 20. Common-shot image panels for shots $X = 4000$ m (a), $X = 6000$ m (b), and $X = 7500$ m (c) in the Marmousi model. Results are clipped with respect to the exact relative impedance perturbation.

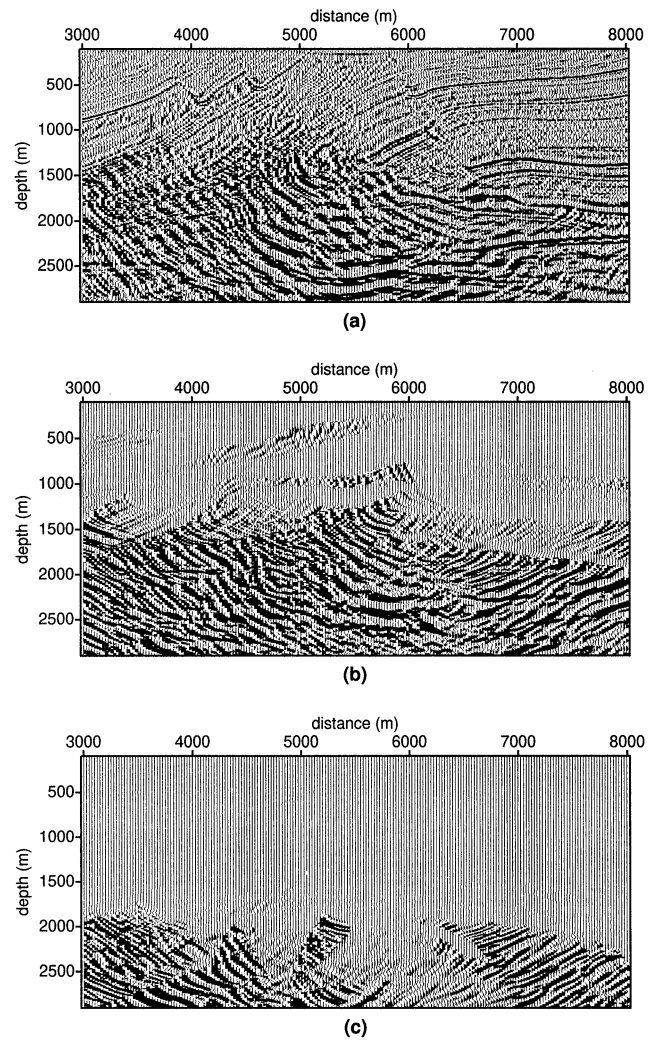


FIG. 21. Common-offset image panels for offsets $h = 200$ m (a), $h = 950$ m (b), and $h = 2575$ m (c) in the Marmousi model. Results are clipped to the exact value of the model. Dynamic artifacts appear.

amplitudes) are specific of migration/inversion, kinematic artifacts concern any migration approach in complex media.

We demonstrate also with synthetic examples that common-diffracting-angle migration/inversion does not give rise to such artifacts. It may provide a convenient tool for methods based on the analysis of CIG [(AVA) (de Hoop et al., 1994) or migration-based velocity analysis (Symes, 1993; Chauris et al., 1999)] in complex media. In noncomplex media, it seems also very promising for AVA analysis (Thierry et al., 1999a) since it directly provide the convenient information, i.e., AVA rather than AVO, avoiding the second migration required by Bleistein (1987) for obtaining the specular reflection angle.

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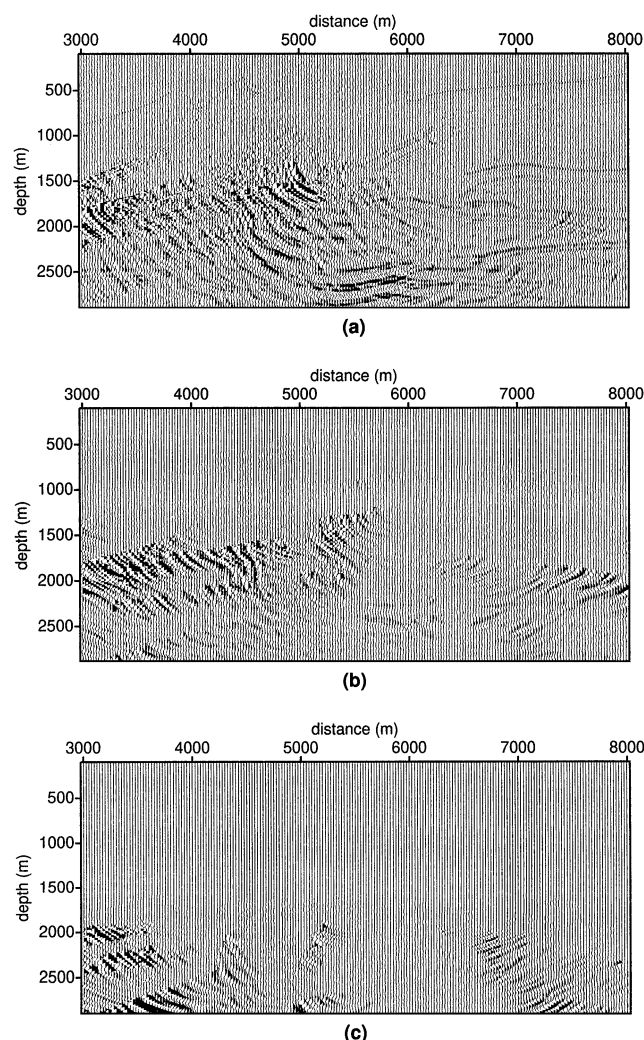


FIG. 22. Common-offset image panels for offsets $h = 200$ m (a), $h = 950$ m (b), and $h = 2575$ m (c) in the Marmousi model. Results are clipped with respect to the maximum value of the plot (percentile 99). Kinematic artifacts appear.

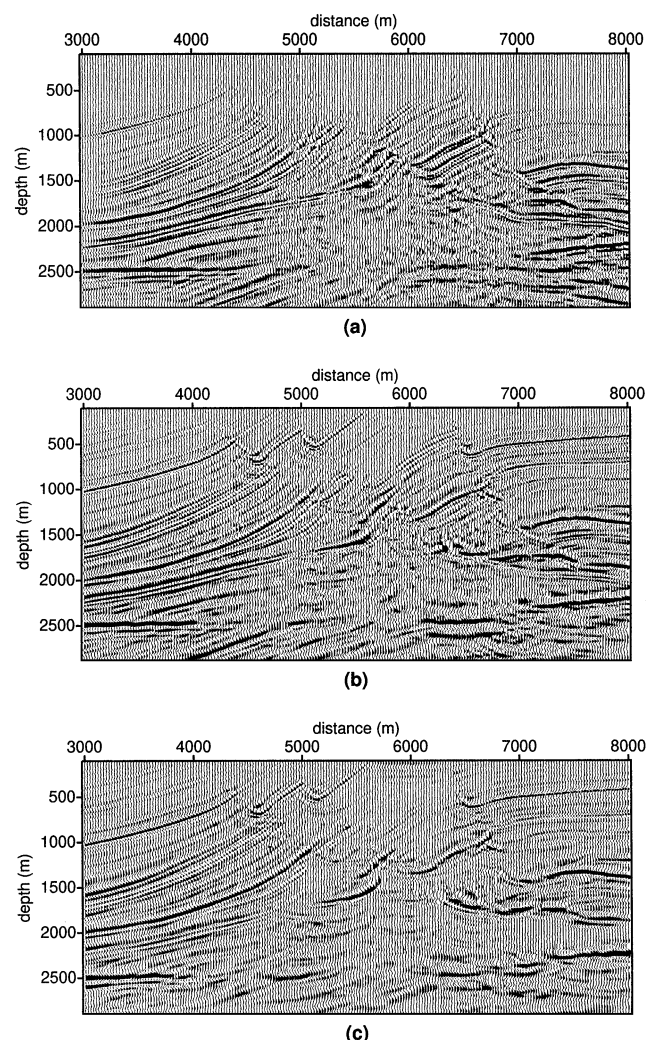


FIG. 23. Common-angle image panels for angles $\theta = 15^\circ$ (a), $\theta = 45^\circ$ (b), and $\theta = 65^\circ$ (c) in the Marmousi model. Results are clipped to the exact value of the model.

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APPENDIX A

LOCAL DIFFERENTIAL REPRESENTATION OF TIC

We demonstrate that local differential condition (13) is equivalent to simplified condition (29). The determinant in equation (13) can be decomposed into two coupled determinants:

$$\det \left| \frac{\partial(\sigma\gamma T p^\sigma)}{\partial(\mathbf{x}\zeta\theta)} \right| = \det \left| \frac{\partial(\sigma\gamma T p^\sigma)}{\partial(\mathbf{x}\sigma\gamma)} \right| \det \left| \frac{\partial(\mathbf{x}\sigma\gamma)}{\partial(\mathbf{x}\zeta\theta)} \right| \quad (\text{A-1})$$

$$= \det \left| \frac{\partial T p^\sigma}{\partial \mathbf{x}} \right|_{\sigma\gamma} \det \left| \frac{\partial \sigma\gamma}{\partial \zeta\theta} \right|_{\mathbf{x}}. \quad (\text{A-2})$$

Consider first the first determinant in equation (A-2). Using

$$p^\sigma = \frac{\partial T}{\partial \sigma} \Big|_{\mathbf{x}\gamma} \quad \text{and} \quad \mathbf{q} = \frac{\partial T}{\partial \mathbf{x}} \Big|_{\sigma\gamma} = 2u \cos\left(\frac{\theta}{2}\right) \begin{pmatrix} \sin(\zeta) \\ \cos(\zeta) \end{pmatrix} \quad (\text{A-3})$$

and changing the order of the partial derivatives, we can obtain

$$\det \left| \frac{\partial T p^\sigma}{\partial \mathbf{x}} \right|_{\sigma\gamma} = \det \left| \mathbf{q}, \frac{\partial \mathbf{q}}{\partial \sigma} \right|_{\mathbf{x}\gamma} = \mathfrak{B}(\mathbf{q}). \quad (\text{A-4})$$

We recognize the Beylkin's determinant (Bleistein, 1987). For the second determinant in equation (A-2), we can do the decomposition

$$\det \left| \frac{\partial \sigma\gamma}{\partial \zeta\theta} \right|_{\mathbf{x}} = \det \left| \frac{\partial \sigma\gamma}{\partial \mathbf{q}} \right|_{\mathbf{x}} \det \left| \frac{\partial \mathbf{q}}{\partial \zeta\theta} \right|. \quad (\text{A-5})$$

Combining expressions (A-4) and (A-5), we get for determinant (A-1),

$$\det \left| \frac{\partial(\sigma\gamma T p^\sigma)}{\partial(\mathbf{x}\zeta\theta)} \right| = -\det \left[\left| \frac{\partial \gamma \sigma}{\partial \mathbf{q}} \right| \Big|_{\mathbf{x}} \mathbf{q}, \frac{\partial \mathbf{q}}{\partial \sigma} \Big|_{\gamma} \right]_{\mathbf{x}} \det \left| \frac{\partial \mathbf{q}}{\partial \zeta\theta} \right| \quad (\text{A-6})$$

$$= -\det \left[\begin{bmatrix} \frac{\partial \gamma}{\partial \mathbf{q}} \Big|_{\mathbf{x}} \mathbf{q} & 0 \\ \frac{\partial \sigma}{\partial \mathbf{q}} \Big|_{\mathbf{x}} \mathbf{q} & 1 \end{bmatrix} \det \left| \frac{\partial \mathbf{q}}{\partial \zeta\theta} \right| \right] \quad (\text{A-7})$$

$$= -\left(\frac{\partial \gamma}{\partial \mathbf{q}} \Big|_{\mathbf{x}} \mathbf{q} \right) \det \left| \frac{\partial \mathbf{q}}{\partial \zeta\theta} \right| \quad (\text{A-8})$$

$$= -\left(\frac{\partial \gamma}{\partial \zeta\theta} \Big|_{\mathbf{x}} \left| \frac{\partial \zeta\theta}{\partial \mathbf{q}} \right| \mathbf{q} \right) \det \left| \frac{\partial \mathbf{q}}{\partial \zeta\theta} \right|. \quad (\text{A-9})$$

Finally, developing Jacobian matrices $\left| \frac{\partial \mathbf{q}}{\partial \zeta\theta} \right|$ and $\left| \frac{\partial \zeta\theta}{\partial \mathbf{q}} \right|$, one can find

$$\left| \frac{\partial \zeta\theta}{\partial \mathbf{q}} \right| \mathbf{q} = \frac{\mathbf{q}^2}{\det \left| \frac{\partial \mathbf{q}}{\partial \zeta\theta} \right|} \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (\text{A-10})$$

and consequently for the local differential version of the TIC the condition,

$$\mathbf{q}^2 \frac{\partial \gamma}{\partial \theta} \Big|_{\mathbf{x}\zeta} \neq 0. \quad (\text{A-11})$$

APPENDIX B

DEMONSTRATION OF CONDITION $\frac{\partial \zeta}{\partial \sigma} \Big|_{\mathbf{x}\gamma} \neq 0$

We demonstrate that the local differential condition (13) implies

$$\frac{\partial \zeta}{\partial \sigma} \Big|_{\mathbf{x}\gamma} \neq 0, \quad (\text{B-1})$$

provided there are no grazing rays and no direct rays.

Let us start with decomposition (A-2):

$$\det \left| \frac{\partial(\sigma\gamma T p^\sigma)}{\partial(\mathbf{x}\zeta\theta)} \right| = \det \left| \frac{\partial T p^\sigma}{\partial \mathbf{x}} \right|_{\sigma\gamma} \det \left| \frac{\partial \sigma\gamma}{\partial \zeta\theta} \right|_{\mathbf{x}}. \quad (\text{B-2})$$

For the common-angle migration, $\det \left| \frac{\partial \sigma \gamma}{\partial \zeta \theta} \right|_{\mathbf{x}} = 1$. For all other cases, we have $\det \left| \frac{\partial \sigma \gamma}{\partial \mathbf{s} \mathbf{r}} \right|_{\mathbf{x}} \neq 0$ and not infinite, and thus we can write using (A-4),

$$\begin{aligned} & \det \left| \frac{\partial (\sigma \gamma T p^\sigma)}{\partial (\mathbf{x} \zeta \theta)} \right| \\ &= \det \left| \mathbf{q}, \frac{\partial \mathbf{q}}{\partial \sigma} \right|_{\mathbf{x} \gamma} \det \left| \frac{\partial \sigma \gamma}{\partial \mathbf{s} \mathbf{r}} \right|_{\mathbf{x}} \left| \frac{\partial \mathbf{s}}{\partial \beta_s} \right|_{\mathbf{x}} \left| \frac{\partial \mathbf{r}}{\partial \beta_r} \right|_{\mathbf{x}}. \end{aligned} \quad (\text{B-3})$$

In the case of no grazing rays, we have

$$\left| \frac{\partial \mathbf{s}}{\partial \beta_s} \right|_{\mathbf{x}} \neq \pm \infty \quad \text{and} \quad \left| \frac{\partial \mathbf{r}}{\partial \beta_r} \right|_{\mathbf{x}} \neq \pm \infty, \quad (\text{B-4})$$

and equation (13) implies (also valid for the common-angle case)

$$\det \left| \mathbf{q}, \frac{\partial \mathbf{q}}{\partial \sigma} \right|_{\mathbf{x} \gamma} \neq 0. \quad (\text{B-5})$$

Using decomposition

$$\frac{\partial \mathbf{q}}{\partial \sigma} \Big|_{\mathbf{x} \gamma} = \frac{\partial \mathbf{q}}{\partial \zeta \theta} \frac{\partial \zeta \theta}{\partial \sigma} \Big|_{\mathbf{x} \gamma} \quad (\text{B-6})$$

and expression of the matrix $\frac{\partial \mathbf{q}}{\partial \zeta \theta}$, we get for condition (B-5),

$$\mathbf{q}^2 \frac{\partial \zeta}{\partial \sigma} \Big|_{\mathbf{x} \gamma} \neq 0. \quad (\text{B-7})$$

Condition (B-7) ensures condition (B-1) if there are no direct rays or grazing rays.